

Statistical Inference

B. Statistical Data Science 2nd Year

Indian Statistical Institute

Class Test - 3

18 November 2025

Total Marks: 10

Name: _____

Roll Number: _____

Instructions

- Answer as much as you can. The maximum you can score is 10.
 - Only write the answer in the space provided or tick the correct option. Any additional writing will not be graded. Rough works should be done on spare pages.
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Question 1. Let $X \sim \text{Normal}(\theta, 4)$, $\theta \in \mathbb{R}$. Consider the problem of testing $H_0 : \theta = 5$ against $H_a : \theta = 8$. What is the probability of Type-I error of the test that rejects H_0 when $X > 6$? [1 mark]

- (a) $\Phi(1)$. (b) $\Phi(-1)$. (c) $\Phi\left(\frac{1}{2}\right)$. (d) $\Phi\left(-\frac{1}{2}\right)$.

Question 2. Let $X \sim \text{Laplace}(\theta, 1)$, $\theta \in \mathbb{R}$. We want to test $H_0 : \theta = -2$ against $H_a : \theta = 0$. What is the probability of Type-II error of the test that rejects H_0 when $X > 2$? [1 mark]

- (a) $1 - \frac{1}{e^2}$. (b) $\frac{1}{e^2}$. (c) $1 - \frac{1}{2e^2}$. (d) $\frac{1}{2e^2}$.

Hint: The pdf of $\text{Laplace}(\theta, 1)$ is $f_\theta(x) = \frac{1}{2}e^{-|x-\theta|}$, $x \in \mathbb{R}$.

Question 3. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} f_\theta(x) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-\frac{x}{\theta}}$, $x \geq 0$. Here, $k > 0$ is known and $\theta > 0$ is the unknown parameter. Which of the following tests is UMP for testing $H_0 : \theta = \theta_0$ against $H_a : \theta < \theta_0$? [1.5 marks]

- (a) Reject H_0 when $\sum_{i=1}^n X_i$ is large. (b) Reject H_0 when $\sum_{i=1}^n X_i$ is small.
(c) Reject H_0 when $X_{(n)}$ is small. (d) Reject H_0 when $X_{(n)}$ is large.

Question 4. Consider $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Uniform}(0, \theta)$, $\theta > 0$, and the problem of testing $H_0 : \theta \geq \theta_0$ against $H_a : \theta < \theta_0$. Define the test function $\varphi(\mathbf{x}) = \mathbf{1}\{x_{(n)} > c_\alpha\}$, with $\mathbb{P}_{\theta_0}(X_{(n)} > c_\alpha) = \alpha$. Which of the following is correct? [1.5 marks]

- (a) φ is UMP level α .
(b) φ is UMP, but its level is not α .
(c) φ is of level α , but it is not UMP.
(d) φ is neither level α , nor UMP.

Question 5. Let $X_1 \sim \text{Normal}(\mu, 5)$ and $X_2 \sim \text{Normal}(\mu, 20)$ be two independent random variables. Which of the following tests is MP for testing $H_0 : \mu = -2$ against $H_a : \mu = 2$? [1.5 marks]

- (a) Reject H_0 if $X_1 + X_2$ is large.
- (b) Reject H_0 if $X_1 + 4X_2$ is large.
- (c) Reject H_0 if $4X_1 + X_2$ is large.
- (d) None.

Question 6. Let $X_1 \sim \text{Exponential}(\theta)$ and $X_2 \sim \text{Exponential}(2\theta)$ be independent, where $\theta > 0$. Which of the following test is UMP for testing $H_0 : \theta = 1$ vs. $H_a : \theta \neq 1$? [1.5 marks]

- (a) Left-tailed test based on $X_1 + 2X_2$.
- (b) Right-tailed test based on $X_1 + 2X_2$.
- (c) Both-tailed test based on $X_1 + 2X_2$.
- (d) UMP test does not exist.

Hint: The pdf of $\text{Exponential}(\theta)$ distribution is $f_\theta(x) = \theta e^{-\theta x}$, $x \geq 0$.

Question 7. Let $X_1, \dots, X_n$ be i.i.d. with pdf $f_\theta(x) = \frac{2}{\theta} x e^{-\frac{x^2}{\theta}}$, $x \geq 0$, where $\theta > 0$. Find a statistic $T(\mathbf{X})$ such that the joint pdf of $X_1, \dots, X_n$ has MLR (increasing) w.r.t. $T(\mathbf{X})$. Derive the UMP test for testing $H_0 : \theta \leq 1$ against $H_a : \theta > 1$. Only write the form of the test. Exact form of the cut-off is not needed. [1+1.5 marks]

Answer:

Question 8. Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p)$, $p \in (0, 1)$. Consider the problem of testing $H_0 : p = p_0$ vs. $H_a : p \neq p_0$. Let $\hat{p}$ be the MLE of p . Identify the likelihood ratio test statistic, the Wald's test statistic and the Rao's score statistic among the following. [1.5 marks]

- (a) $\frac{n(\hat{p} - p_0)^2}{\hat{p}(1 - \hat{p})}$.
- (b) $\frac{n(\hat{p} - p_0)^2}{p_0(1 - p_0)}$.
- (c) $2n \left\{ \hat{p} \log \left(\frac{\hat{p}}{p_0} \right) + (1 - \hat{p}) \log \left(\frac{1 - \hat{p}}{1 - p_0} \right) \right\}$.